

MA1513 Exam

AY2024/2025 Semester I

MCQ/FIB (Q1 – Q15; 3 marks each)

1. Given $\mathbf{Ax} = \mathbf{b}$ is a non-homogeneous linear system where $\mathbf{A}$ is a 3×5 matrix. Suppose the

general solution is given by
$$\begin{cases} x_1 = s + 2t \\ x_2 = t - s - 1 \\ x_3 = s \\ x_4 = t + 2 \\ x_5 = t \end{cases}.$$

Which of the following statements are correct? (Pick all correct options)

- The RREF of the augmented matrix $(\mathbf{A} \mid \mathbf{b})$ has exactly 2 non-pivot columns.
 - Any REF of the matrix $\mathbf{A}$ does not have a zero row.
 - $\begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \\ 1 \end{pmatrix}$ is a solution of $\mathbf{Ax} = \mathbf{0}$.
 - $\begin{pmatrix} 0 \\ 1 \\ 0 \\ -2 \\ 0 \end{pmatrix}$ is a solution of $\mathbf{Ax} = \mathbf{b}$.
 - For any 3×1 column matrix $\mathbf{c}$, the linear system $\mathbf{Ax} = \mathbf{c}$ is consistent.
2. Suppose $\mathbf{A}$ is a non-singular $n \times n$ matrix. Which of the following statements are true? (Pick all correct options)
- Any REF of $\mathbf{A}^T$ is non-singular.
 - $\mathbf{A}$ has n linearly independent eigenvectors.
 - The row space of $\mathbf{A}$ is $\mathbb{R}^n$.
 - The rank of $\mathbf{A}^k$ is n for any integer $k > 1$.
 - We can express any column of $\mathbf{A}$ as a linear combination of the remaining columns of $\mathbf{A}$.
3. Suppose $\mathbf{A}$ is a 2×2 matrix with determinant 2. Then
- $\det(-\mathbf{A}) = a$
 - $\det(2\mathbf{A}^T\mathbf{A}) = b$
 - $\det(4\mathbf{A}^{-1}) = c$
- where $a = \underline{\hspace{1cm}}$, $b = \underline{\hspace{1cm}}$, $c = \underline{\hspace{1cm}}$. (Fill in the blanks)
4. Which of the following vectors are coplanar with $(1, 2, 3)$ and $(3, 2, 1)$? (Pick all correct answers.)
- $(1, 3, 2)$
 - Any linear combination of $(1, 2, 3)$ and $(2, 1, 3)$
 - $(2, 1, 0)$
 - $(2, 0, 2)$
 - Any linear combination of $(1, 0, -1)$ and $(0, 1, 2)$
5. Which of the following is a basis for the subspace given by $\text{span}\{(1, 0, 1, 1), (1, 2, 0, 1), (1, 4, -1, 1), (1, -2, 2, 1), (2, 2, 1, 1)\}$? (Pick all correct answers)

- i. $\{ (1, 0, 1, 1), (1, 2, 0, 1), (1, 4, -1, 1) \}$
 - ii. $\{ (1, 4, -1, 1), (1, -2, 2, 1), (2, 2, 1, 1) \}$
 - iii. $\{ (1, 0, 0, 0), (0, 1, 0, 0), (0, 0, 1, 0) \}$
 - iv. $\{ (1, 2, 0, 1), (1, 4, -1, 1), (1, -2, 2, 1) \}$
 - v. $\{ (1, 0, 1, 1), (1, 4, -1, 1), (2, 2, 1, 1) \}$

6. The coordinate vector of a vector $\mathbf{v}$ with respect to the basis $S = \{ (1, 0, 1), (0, 1, 1), (1, 1, 0) \}$ is given by $(\mathbf{v})_S = (3, 6, 9)$. Then the coordinate vector of $\mathbf{v}$ with respect to the basis $T = \{ (2, 1, 0), (0, 2, 1), (1, 0, 2) \}$ is given by $(\mathbf{v})_T = (a, b, c)$ where $a = \underline{\hspace{1cm}}$, $b = \underline{\hspace{1cm}}$, $c = \underline{\hspace{1cm}}$ (FIB)

7. Which of the following sets are linearly independent? (Pick all correct options)
 - i. $\{ (10, 20, 30, 40), (11, 22, 33, 44) \}$
 - ii. $\{ (1, 5, 1, 3), (2, 0, 2, 4), (0, 0, 0, 0) \}$
 - iii. $\{ (1, 1, 0, 0), (0, 0, 1, 1), (1, 0, 1, 0), (0, 1, 0, 1) \}$
 - iv. $\{ (1, 2, 3, 0), (2, 3, 1, 0), (3, 1, 2, 0) \}$
 - v. $\{ (1, 1, 1, 1), (2, 2, 0, 2), (3, 0, 3, 0), (0, 0, 0, 4) \}$

8. Three of the following subspaces of $\mathbf{R}^3$ are the same as each other. Select the two subspaces that are not the same as the three.
 - i. $\text{Span}\{ (1, -1, 0), (1, 0, -1) \}$
 - ii. Nullspace of $\begin{pmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \end{pmatrix}$
 - iii. Row space of $\begin{pmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ 1 & 1 & 1 \end{pmatrix}$
 - iv. Column space of $\begin{pmatrix} 2 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & -1 & 0 \end{pmatrix}$
 - v. Solution space of $x + 2y + 3z = 0$

9. Let $T: \mathbf{R}^3 \rightarrow \mathbf{R}^3$ be a linear transformation such that $T\left(\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}\right) = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$ and $T\left(\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}\right) = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$.
 Then $T\left(\begin{pmatrix} 2 \\ 5 \\ 3 \end{pmatrix}\right) = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$ where $a = \underline{\hspace{1cm}}$, $b = \underline{\hspace{1cm}}$, $c = \underline{\hspace{1cm}}$ (FIB)

10. Given $\mathbf{A}$ is a 3×3 diagonalizable matrix with eigenvalues 1 and 3 with corresponding eigenspaces $E_1 = \text{span}\left\{\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}\right\}$ and $E_3 = \text{span}\left\{\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}\right\}$
 Which of the following matrices diagonalize $\mathbf{A}$? (Pick all correct answers)
 - i. $\begin{pmatrix} 0 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{pmatrix}$
 - ii. $\begin{pmatrix} 2 & 4 & 0 \\ 0 & 4 & 0 \\ 2 & 4 & 6 \end{pmatrix}$

- iii. $\begin{pmatrix} 1 & 1 & 2 \\ 1 & 0 & 2 \\ 1 & 1 & 2 \end{pmatrix}$
- iv. $\begin{pmatrix} 2 & 1 & 0 \\ 1 & 0 & 0 \\ 2 & 1 & 1 \end{pmatrix}$
- v. $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 1 & 0 \end{pmatrix}$

11. Given a 4x4 matrix $\mathbf{A}$ with characteristic polynomial $x(x-1)^a(x+1)^b$ for some positive integers a and b . Which of the following statements are true? (Pick all correct answers)

- i. The nullity of $\mathbf{A}$ is 1.
- ii. The dimension of one of the eigenspaces of $\mathbf{A}$ is 2.
- iii. Exactly one of the eigenvalues of $\mathbf{A}$ has multiplicity 2.
- iv. $\mathbf{A}$ cannot be diagonalizable.
- v. $\det(\mathbf{A}) = 0$.

12. Given a 3 x 3 SDE $\mathbf{y}' = \mathbf{A}\mathbf{y}$ with fundamental set of solutions

$$\mathbf{x}_1 = e^t \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}, \mathbf{x}_2 = e^{-t} \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \mathbf{x}_3 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}. \text{ Supposed } \mathbf{y}(0) = \begin{pmatrix} 3 \\ 6 \\ 4 \end{pmatrix}. \text{ Then the particular}$$

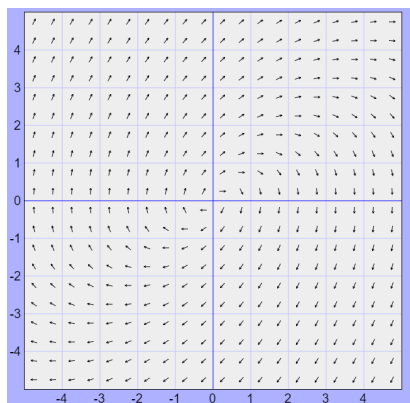
solution of the initial value problem is

$$\mathbf{y} = a\mathbf{x}_1 + b\mathbf{x}_2 + c\mathbf{x}_3 \text{ where } a = _, b = _, c = _ \text{ (FIB)}$$

13. An SDE is given by $\begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \begin{pmatrix} 0 & 2 \\ -2 & -4 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$. Which of the following are solutions of the SDE?

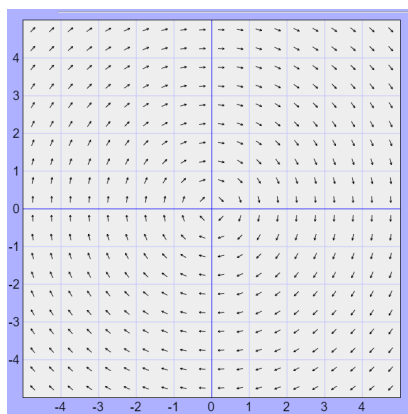
- i. $\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = e^{-2t} \begin{pmatrix} 1/2 \\ -1/2 \end{pmatrix}$
- ii. $\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = e^{-2t} \begin{pmatrix} -1/2 \\ 0 \end{pmatrix}$
- iii. $\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = e^{-2t} \begin{pmatrix} -t-1 \\ t \end{pmatrix}$
- iv. $\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = e^{-2t} \begin{pmatrix} -t \\ t - \frac{1}{2} \end{pmatrix}$
- v. $\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = e^{-2t} \begin{pmatrix} t - \frac{1}{2} \\ -t \end{pmatrix}$

14. Which of the following are the possible direction fields for an SDE $\mathbf{y}' = \begin{pmatrix} 0 & a \\ -1 & b \end{pmatrix} \mathbf{y}$ for some a and b ?



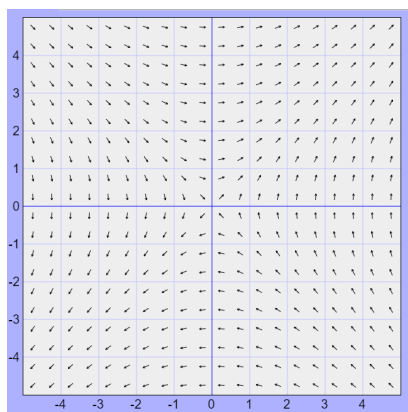
- i. e.g. $\begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix}$

ii.

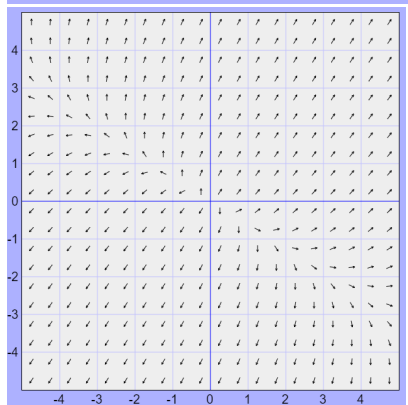


e.g. $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$

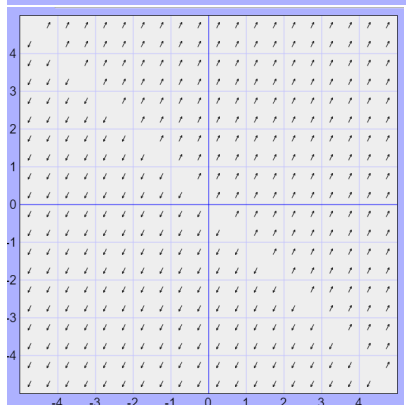
iii.



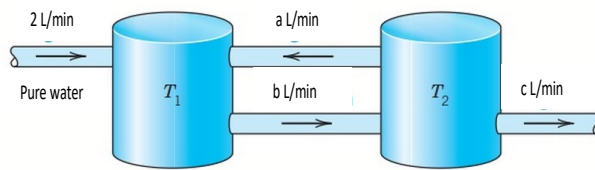
iv.



v.



15. Consider the two tanks in the diagram. Tank 1 holds 100 litre of salt water and tank 2 holds 200 litre of salt water.



Suppose the amount of salt in tank 1 and 2 are $y_1(t)$ and $y_2(t)$ respectively and the SDE that describe the amount of salt in the two tanks is given by $\begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \frac{1}{100} \begin{pmatrix} -6 & 2 \\ 6 & -3 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$.

Then $a = \underline{\hspace{1cm}}$, $b = \underline{\hspace{1cm}}$, $c = \underline{\hspace{1cm}}$ (FIB).

Written Questions (Q16 – Q19)

Q16. [10 marks]

A manufacturer makes four types of furniture A, B, C, D. The company has available 2600 units of wood, 600 units of upholstery and 2400 units of labour. The manufacturer wants a production schedule that uses all of these resources. The various products require the following amounts of resources.

	A	B	C	D
Wood	4	4	3	4
Upholstery	0	1	0	2
Labour	2	3	4	4

- Set up a linear system with variables a, b, c, d , which represent the number of pieces of products A, B, C, D respectively to be manufactured.
- Solve the linear system in (i).
- What is the range for the possible values of d ?

Q17. [10 marks]

Let $A = \begin{pmatrix} 1 & 2 & 1 & 0 \\ 2 & 4 & 0 & -2 \\ 1 & 2 & 2 & 1 \end{pmatrix}$ and $b = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$.

- Find a basis for and the dimension of the column space of A .
- Find the projection of b onto the column space of A .
- Find a unit vector in $\mathbb{R}^3$ which is orthogonal to all the vectors in the column space of A .

Q18. [12 marks]

A certain species of animals has two life stages: juvenile (up to 1 year old) and adult. Each year,

- an adult produces an average of 1.2 juveniles;
- 50% of the juveniles survive to become adults;
- 40% of the adults survive.

Let $x_n = \begin{pmatrix} j_n \\ a_n \end{pmatrix}$ where j_n and a_n are the number of juveniles and adults respectively in year n .

- Find the stage matrix A such that $x_{n+1} = Ax_n$.
- Diagonalize the matrix A^n .
- Suppose the initial populations j_0 and a_0 for juveniles and adults are respectively 10,000 and 8,000. Will the populations be stabilized in the long term? Why?
- True or false: Regardless of the initial populations j_0 and a_0 , the ratio of juvenile to adult populations will stabilize in the long term. Justify your answer.

Q19. [13 marks]

A particular solution of an SDE $y' = Ay$ is given by: $y = 3e^{-t} \begin{pmatrix} -\cos t + \sin t \\ 2 \cos t \end{pmatrix} - 3e^{-t} \begin{pmatrix} -\sin t - \cos t \\ 2 \sin t \end{pmatrix}$.

- Find the matrix A .
- State the type and stability of the equilibrium point.
- Sketch the trajectory of the given particular solution on the phase plane, indicating on the plane the initial value when $t = 0$.
- What is the tangent vector of the trajectory at the point when it first crosses the horizontal y_1 -axis after $t = 0$?